

SIMATS ENGINEERING

ASSESSMENT TOOL 3

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COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 35%

Dr

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5

8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I K. Lokesh Chandra (192425358) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Lokesh Chandra

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

ANSWERS

DATABASE MANAGEMENT SYSTEMS TECHNICAL PRESENTATION

Assessment Tool 3

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INTRODUCTION

Definition

A **Database Management System (DBMS)** is software that allows users to create, store, retrieve, update, and manage data efficiently. It acts as an interface between the database and users, ensuring that data is organized, secure, and easily accessible.

Key Features

Stores and manages large amounts of data.

Supports multiple users simultaneously.

Provides data security and privacy.

Reduces data redundancy.

OBJECTIVES

The main objectives of a Database Management System are:

Reduce data redundancy by avoiding duplicate data.

Improve data consistency across the database.

Provide secure access using authentication and authorization.

Support multiple users through concurrency control.

Maintain data integrity and accuracy.

Provide backup and recovery facilities.

Enable efficient storage and quick retrieval of data.

Simplify database administration and maintenance.

Benefits

Faster data processing

Better decision making

ADVANTAGES

A DBMS offers several benefits over traditional file systems.

Advantages

Minimizes data redundancy.

Provides better data security.

Supports concurrent access by multiple users.

Maintains data integrity and consistency.

Enables efficient backup and recovery.

Allows quick searching and retrieval of information.

Simplifies application development.

Improves data sharing among departments.

REAL-WORLD APPLICATIONS

Applications of DBMS

DBMS is widely used in many industries.

Banking

Customer account management

ATM transactions

Online banking

Healthcare

Electronic medical records

Appointment scheduling

Hospital management

E-Commerce

Product inventory

Online orders

Payment processing

Government

Citizen records

Tax management

Passport services

Q2: FILE SYSTEM VS DBMS

File-Based System	DBMS
Data is stored in separate files.	Data is stored in a centralized database.
Data duplication is common.	Reduces data redundancy.
Less secure.	Provides better security.
Difficult to update data.	Easy to update and manage data.
Backup is difficult.	Easy backup and recovery.

Real-Life Examples

Banking

File System: Customer details are stored in different files, causing duplicate records.

DBMS: All customer information is stored in one database, making transactions faster and more secure.

Q3: ANSI/SPARC ARCHITECTURE

The ANSI/SPARC architecture divides the database into **three levels** for easy data management.

Three Levels

External Level: Shows data to users based on their needs.

Conceptual Level: Describes the complete structure of the database.

Internal Level: Shows how data is stored in the computer.

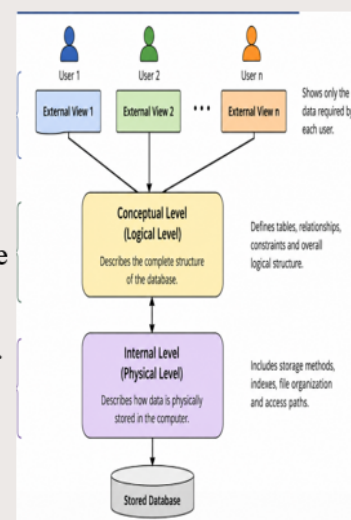
Advantages

Provides data security.

Makes database management easier.

Supports logical and physical data independence.

Improves flexibility and maintenance.



Q4: DATA MODELS

1. Hierarchical Model

Data is organized in a tree structure.

Each child has only one parent.

Example: Organization chart.

2. Network Model

Data is connected using many-to-many relationships.

A child can have multiple parents.

Example: Railway reservation system.

3. Relational Model

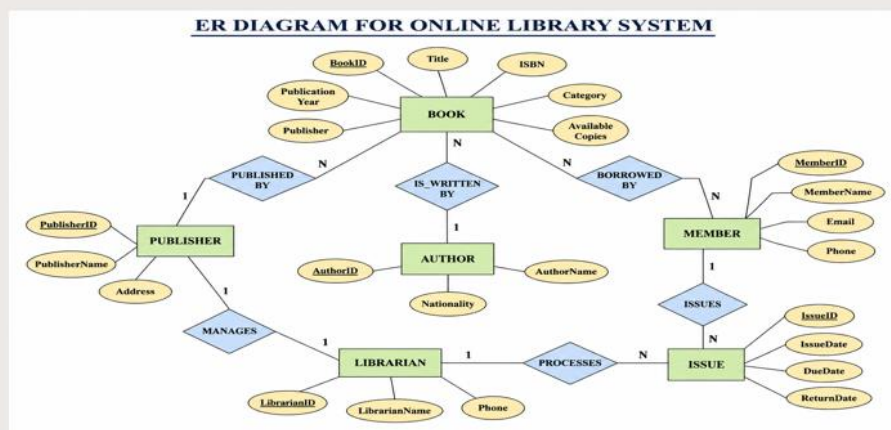
Data is stored in tables (rows and columns).

Tables are connected using keys.

Example: Banking and Student Management System.

Q5: ER DIAGRAM

- Entities: Book, Member, Librarian. This concept is important in database management and is widely used in real-world applications.



Q6: EER MODEL

- **Enhanced Entity-Relationship (EER) Model**
- The **EER Model** is an extension of the ER Model that represents complex relationships in a database.
- **Key Concepts**
- **Generalization:** Combines similar entities into one common entity.
- **Specialization:** Divides one entity into different sub-entities.
- **University Example**
 - Person (Superclass)
 - Student
 - Faculty
 - Staff
- **Advantages**
- Easy to represent complex data.
- Improves database design.

Q7: DATA INDEPENDENCE

Data Independence means changes in the database do not affect the application programs.

Types of Data Independence

1. Logical Data Independence

Changes in the logical structure do not affect users or applications.

2. Physical Data Independence

Changes in the physical storage do not affect the logical structure.

Advantages

Easy database maintenance.

Improves flexibility.

Saves time and cost.

Makes the database more reliable.

Q8: DBMS COMPONENTS

- A **DBMS** consists of different software components that help manage data efficiently.
- **Main Components**
- **Query Processor:** Executes SQL queries and retrieves data.
- **Storage Manager:** Stores and manages data in the database.
- **Transaction Manager:** Ensures safe and accurate execution of transactions.
- **Buffer Manager:** Manages data between memory and disk.
- **File Manager:** Organizes and maintains database files.
- **Advantages**
 - Fast data processing.
 - Secure data management.
 - Reliable transaction handling.

Q9: STRONG VS WEAK ENTITY

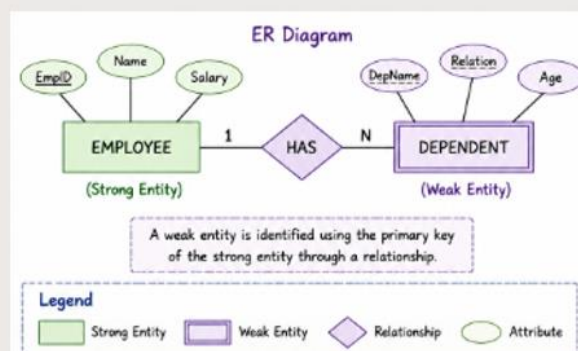
An **Entity** is an object or person that stores information in a database.

Strong Entity

- Has its own **Primary Key**.
- Can exist independently.
- **Example:** Employee, Student.

Weak Entity

- Does not have its own Primary Key.
- Depends on a Strong Entity.
- **Example:** Dependent of an Employee.



Q10: RETAIL CHAIN EER - SLIDE 2

Main Entities

- Customer
- Product
- Store
- Order
- Payment

Advantages

- Organizes retail data efficiently.
- Reduces data redundancy.
- Improves order and inventory management.
- Makes database design easy to understand.

Legend



Relationships

- Customer places an Order.
- Store sells Products.
- Order contains Products.
- Customer makes Payment.

